

The Guardian: A Hybrid Deep Learning and Heuristic Framework for Real-Time Social Media Scam Detection

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Abstract—The rapid growth of social media platforms has significantly increased users’ exposure to online scams, including phishing, fraudulent promotions, and malicious links. Existing detection approaches primarily rely on either URL-based heuristics or standalone machine learning models, which often fail to capture both structural and contextual indicators of scams, particularly in real-time environments. To address these limitations, this paper presents The Guardian, a hybrid scam detection framework that integrates heuristic URL analysis with deep learning-based contextual understanding for real-time social media protection. The proposed system employs a dual-layer architecture consisting of a lightweight heuristic engine for rapid URL-based risk assessment and a DistilBERT-based model for context-aware text classification. A maximum-risk fusion strategy is introduced to combine outputs from both components, ensuring that critical risk signals are preserved. Additionally, the framework incorporates a Safe Preview mechanism that enables users to inspect potentially malicious links within a controlled, sandboxed environment, thereby minimizing exposure risk. The system is implemented as a browser extension supported by a FastAPI backend, enabling real-time detection and user feedback during browsing sessions. Experimental evaluation demonstrates that the hybrid approach significantly improves detection robustness compared to single-method systems, particularly in scenarios involving obfuscated or context-dependent scams. The proposed framework provides a scalable, practical, and user-centric solution for enhancing security in modern social media environments.

Index Terms—Cybersecurity, Social Media Scams, Deep Learning, DistilBERT, Heuristic Analysis, Real-time Detection.

I. INTRODUCTION

The fast growth of social media sites has altered how people interact, exchange information and how they transact online. Nevertheless, it has also resulted in a substantial rise in online

scams, such as phishing, fraudulent promotions, impersonation and malicious URL-based exploits. These scams are frequently based on technical manipulation and social engineering tricks, making them increasingly difficult to detect. Recent research points out that not only malicious links, but contextual clues in user-generated content are used by attackers to enhance the success of these attacks [1], [3].

Conventional scam detection techniques are mainly based on URL-based heuristics or blacklist systems which are based on the detection of known bad domains and patterns. Although these techniques are quick and effective in detecting URLs, they do not adequately detect newly minted or obfuscated URLs [6], [7]. Conversely, machine learning and deep learning-based methods, especially those that use transformer-based semantics like BERT and its derivatives, have been shown to perform quite well at comprehending text much better in contextual (e.g., phishing or scam-related) text [1], [2]. These models, though effective, are usually computationally intensive and might be unsuitable for real-time deployment unless optimized.

More recent studies have investigated hybrid detection processes that integrate heuristic and machine learning techniques to enhance detection strengths [11], [13], [15]. These systems take advantage of the accuracy and speed of rule-based solutions and the context-aware capabilities that deep-learning models offer. Nevertheless, available hybrid solutions can be restricted to offline analysis or not smoothly blended into real-time user environments like web browsers. Moreover, there are numerous methods that lack user-centered safety measures and actively block communication with potentially harmful materials.

Moreover, the growing complexity of scams shows the relevance of context-aware detection. Attacks based on social engineering often depend on the use of persuasive text, urgency signals, and misleading stories within the text, and such messages cannot be effectively analyzed by their URLs alone [16], [18]. This requires systems that can effectively analyze both structural and contextual features within a single framework.

In order to overcome these limitations, this paper presents *The Guardian*, a hybrid real-time scam detection system that combines the heuristic analysis of the URL with deep learning-based contextual classification. The proposed system uses a dual-layer architecture built upon a lightweight URL analysis engine and a DistilBERT-based text classification model, allowing for the precise identification of explicit and implicit scam indicators. Moreover, a Safe Preview feature is presented, in which users have the opportunity to access suspicious URLs in a controlled sandbox environment, thus minimizing exposure risk. This work bridges the gap between theoretical detection models and practical, deployable cybersecurity solutions by integrating real-time processing, hybrid intelligence, and user-centric protection mechanisms within a unified framework.

This work has its primary contributions as follows:

- 1) A hybrid detection framework combining heuristic URL analysis with transformer-based contextual understanding to enhance scam detection accuracy.
- 2) A real-time, browser-integrated system that spots scams when a user accesses web content.
- 3) An innovative fusion approach to merging the results of heuristic and deep learning elements to improve detection strength.
- 4) Safe Preview Mechanism: an extra safeguard to the user of the device by inspection of the links.
- 5) End-to-end implementation that displays the feasibility of the proposed approach in a real-world situation.

The rest of this paper is structured in the following manner. Section II has a literature review of related studies on phishing and scam detection. Section III contains the methodology proposed. Section IV describes the system architecture and implementation. Section V gives the experiment outcomes and assessment. Lastly, Section VI wraps up the paper and provides future directions of research.

II. RELATED WORK

Detection of online scams and phishing has been a research topic in a variety of fields, including machine learning, natural language processing, and web security. Existing approaches can be broadly categorized into URL-based detection methods, deep learning-based textual analysis, hybrid frameworks, and real-time browser-based systems.

A. URL-Based Phishing Detection

Although early and more recent phishing detection methods heavily depend on structural characteristics of URLs, i.e. domain characteristics, lexical patterns and hosting information.

Handcrafted URL features have been shown to be effective in training machine learning models used to detect known phishing patterns [6], [7]. Ensemble-based techniques also enhance the accuracy of detection with the combination of a set of classifiers and features [10].

Although effective, URL-based methods have natural shortcomings in addressing newly created, obfuscated or short-lived malicious links. Shortening of URLs, domain generation algorithms and homoglyph attacks are among the techniques that attackers often use to avoid detection. This leads to the fact that purely URL-based systems tend to have lower generalization capabilities when operating in dynamic threat environments.

B. Deep Learning for Scam and Phishing Detection

As natural language processing has advanced, deep learning models have been utilized more to detect scams. Transformer-based models like BERT have demonstrated high potential in building contextual semantics and detecting phishing or fraudulent patterns of language use [1]. Light versions like DistilBERT are an alternative that is computationally efficient yet perform competitively [2].

Recent research has also used deep neural networks and NLP to identify phishing content in emails, SMS, and posts on social media, showing better accuracy than the old methods of machine learning [3], [5]. The models are especially useful in detecting context-dependent scams that are based on persuasive language, urgency indicators, and social engineering tricks.

Nevertheless, deep learning models can be computationally intensive and introduce latency, which is unsuitable for real-time deployment without optimization. Moreover, they are restricted from using text information only, thus diminishing their potential to detect structural signals like malicious URLs.

C. Hybrid Detection Approaches

In a bid to eliminate the weaknesses of single methods, hybrid techniques involving machine learning and heuristic methods have been suggested. These architectures take advantage of the speed and accuracy of rule-based detection and the generalization aspect of machine learning models [11], [12]. Multi-layered detection systems also increase strength as they combine several sources of features and detection policies [14].

New hybrid models with textual and URL-based features have proven to be more effective in identifying phishing and scam-based material [13], [15]. These methods offer a better understanding of malicious behavior by integrating structural and contextual signals.

However, the majority of existing hybrid systems are oriented towards enhancing the performance of the detection and are typically tested in offline contexts. There has been little focus on real-time deployment, system integration and user interactive safety mechanisms.

D. Context Aware and Social Engineering Detection

The techniques of social engineering in scams are becoming more popular among modern scammers because they exploit human psychology, using contextual inferences in text. Studies on this topic highlight the need to have context-sensitive detection systems that can determine intent, sentiment and behavioral patterns [16], [18].

Context-sensitive models have been demonstrated to identify fine and indirect signs of scams that are not able to be detected using traditional rule-based systems. Nevertheless, they are normally used independently and are not integrated with structural analysis techniques like URL analysis.

E. Real-Time and Browser-Based Security Systems

Real-time detection systems, especially those that are put in place as browser-based solutions, are instrumental in safeguarding users as they engage in active browsing. Previous research investigated client-side and browser-based solutions to malicious web content detection and phishing attacks [17], [19]. These systems give instant feedback to users and lessen exposure to harmful contents.

Nevertheless, the current browser-based solutions tend to be based on blacklists, or lightweight heuristics, which are not able to identify advanced or context-specific scams. Moreover, most systems do not have more sophisticated user protection systems like regulated interaction block with suspicious links.

F. Research Gap

Based on the literature reviewed, one can identify a few limitations:

- 1) Current URL-based approaches have no context insight and are vulnerable to obfuscated or new attacks.
- 2) Deep learning techniques offer good contextual information but are computationally intensive and cannot be implemented in real-time in many situations.
- 3) Hybrid systems enhance the accuracy of detectors but are not commonly deployed in real-time and user-facing systems.
- 4) The existing solutions do not have user-friendly safety features to prevent access to malicious content.

G. Positioning of This Work

In order to overcome these drawbacks, we suggest a hybrid real-time scam detection model which incorporates heuristic analysis of URLs and transformer-based contextual knowledge. To increase user safety, the proposed system integrates structural and contextual detection within a browser-integrated environment, unlike the current approaches, and provides a Safe Preview mechanism. This integrated methodology can be used to not only detect the risks but also preemptively mitigate them, bridging the gap between research models and actual deployment.

III. PROPOSED METHODOLOGY

This section presents the proposed hybrid system for real-time social media scam detection. The methodology is designed to integrate heuristic URL analysis and deep learning contextual understanding while maintaining a strong connection to practical system implementation.

A. General Overview of the Proposed Approach

The input to the system is in the form of textual context and a URL:

$$I = (T, U) \quad (1)$$

where T is the textual content that is extracted and U is the URL. There are two independent analysis elements that produce risk scores:

- S_{url} : heuristic URL based risk score.
- S_{text} : the probability of a contextual scam score.

These scores are combined using a maximum-risk fusion strategy to produce the final risk score (R).

B. Heuristic URL Analysis (URL Engine)

Rule-based analysis is done by the URL engine to identify structural signs of malicious URLs in a minimal amount of latency.

1) *Feature Extraction*: The following features are obtained:

- Suspicious keywords (e.g., “login”, “verify”, “free”)
- Length and complexity of URLs.
- Use of IP addresses instead of domain names
- Availability of URL shorteners.
- Abnormal domain patterns

2) *Heuristic Scoring*: Each triggered rule contributes to a cumulative risk score:

$$S_{url} = \sum w_i \quad (2)$$

where w_i is the weight of the corresponding feature of the detected heuristic.

C. Text Analysis (AI Engine)

The AI engine analyzes textual data with an optimized DistilBERT model in order to detect scam intent.

1) *Preprocessing*:

- Normalize text
- Replace URLs with a placeholder token.
- Tokenize input to process model.

2) *Model Output*: The output of the model is a probability score:

$$S_{text} \in [0, 1] \quad (3)$$

representing the probability that the content is of scam nature.

D. Novel Strategy of Hybrid Fusion

The system uses a maximum-risk fusion strategy to integrate structural and contextual signals:

$$R = \max(S_{url}, S_{text}) \quad (4)$$

This method will not dilute the important indicators because high-risk indicators of either of the components are retained.

E. Real-Time Detection Pipeline

The detection workflow is:

- 1) URLs and other text around web content extraction.
- 2) Sending data to the back-end API.
- 3) Parallel processing of analysis modules (URL and text).
- 4) Combination of results to obtain final risk score.
- 5) Output of classification results to browser extension.

F. Safe Preview Mechanism

To minimize exposure of users to harmful materials, the system includes a Safe Preview mechanism:

- 1) Metadata of the target URL.
- 2) Rendering of the page without executing the scripts.
- 3) Sandboxed preview on an isolated environment (optional).

G. Output Representation

The system provides a formal output which contains:

- Risk score (R) (scaled to 0–100)
- Risk classification (Safe, Suspicious, Dangerous)
- Determined indicators that led to the decision.

H. Summary

The proposed methodology combines both the heuristic and deep learning methods into one framework that is consistent with the real-life application. The system allows the detection of social media scams with the appropriate accuracy and in real-time by integrating a quick structural analysis with contextual information and a strong fusion strategy.

IV. SYSTEM ARCHITECTURE

This section outlines the architecture of *The Guardian*, which is a modular, hybrid system to detect scams in web environments in real-time. The architecture combines both client-side data mining and backend intelligence processing, which allows easy and scalable deployment.

A. Architectural Overview

The system is based on a client-server architecture, which is composed of three main layers:

- 1) Client Layer (Web Browser Extension)
- 2) Backend Processing Layer (FastAPI Server)
- 3) Detection Engine Layer (Multi-purpose Analysis Modules)

On a big picture, the system works by parsing user-facing content on request and sends the appropriate data to the backend and displays to the browser a structured risk analysis. The architecture illustrates the end-to-end interaction between the browser extension and the backend detection system. The client extracts contextual data and URLs, which are processed by parallel detection modules. The fusion engine integrates outputs from both heuristic and AI-based analysis to generate a final risk assessment, ensuring efficient and real-time scam detection.

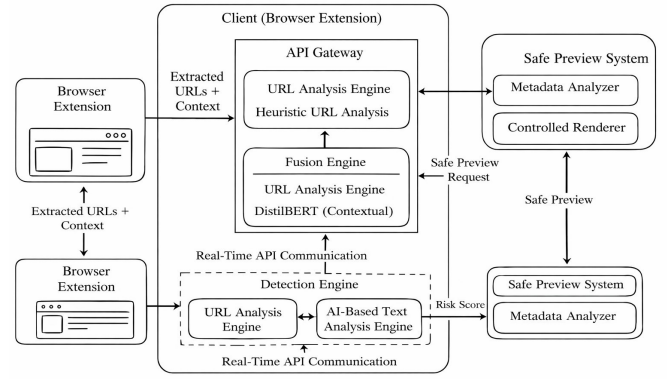


Fig. 1. System architecture of The Guardian, illustrating the interaction between the browser extension and the hybrid detection backend.

B. Client Layer: Browser Extension

The client layer is a browser extension that is in charge of real-time communication with web content.

1) *Content Extraction Module*: This module scans web pages and finds hyperlinks (URLs), surrounding textual context, and content which is visible to the user and related to possible scams. The data extracted is in the form of:

$$I = (T, U) \quad (5)$$

in which T is contextual text and U is the URL.

2) *Communication Interface*: The extension communicates with the backend via RESTful API calls. Data is extracted and sent to the server to be analyzed and responses are received in the form of a JSON with risk scores and classifications.

3) *User Interface Module*: The extension gives visual feedback in real-time by:

- Color-coded signs (Safe / Suspicious / Dangerous)
- Badges on links.
- Safe Preview Interactive prompts.

This ensures that users are informed of potential risks in real-time without disrupting their browsing experience.

C. Backend Processing Layer

The application is written in FastAPI, and it is the main processing unit of the system.

1) *API Gateway*: The backend has the following endpoints:

- `/scan/main` – Key endpoint of detection.
- `/safe-preview` – Safe-preview processing.
- `/proxy` – Controlled content rendering.

These endpoints receive incoming requests and coordinate processing among system components.

2) *Request Processing Pipeline*: When input data is received the backend does input validation and normalization, routing to detection modules, aggregation of results, and response generation. This pipeline is designed to work with low-latency processing to enable real-time detecting.

D. Detection Engine Layer

The detection engine is the analytical part of the system, which consists of two parallel modules.

1) *URL Analysis Engine*: This module does heuristic analysis of URLs based on predefined rules and feature extraction methods. It detects structural risk indicators like suspicious domains, abnormal patterns, and familiar phishing traits. The result is a risk score:

$$S_{url} \quad (6)$$

2) *Text Analysis Engine with AI*: The module is based on a highly-tuned DistilBERT model that analyzes text in context and detects scam-intent. It takes in tokenized input and generates a score of probability:

$$S_{text} \quad (7)$$

3) *Fusion Engine*: The fusion strategy of the methodology combines the outputs of the two engines:

$$R = \max(S_{url}, S_{text}) \quad (8)$$

This strategy prioritizes the highest risk signal, ensuring that strong indicators from either structural or contextual analysis are not suppressed during the final risk assessment.

E. Safe Preview Subsystem

The Safe preview subsystem is a secure method of checking suspicious links without subjecting the user to possible dangers. This subsystem comprises of:

- 1) **Metadata Analyzer**: Get page metadata (title, domain, and headers).
- 2) **Static Renderer**: Shows page content without running scripts.
- 3) **Sandbox Environment**: Isolates full rendering in a controlled environment.

F. Data Flow and Interaction

The workflow of the end-to-end system is as following: 1) User visits a web page; 2) URL and context are extracted; 3) Information is posted to backend API; 4) Parallel analysis is done; 5) Results are combined for a final risk score; 6) Response is returned; 7) Risk indicators are displayed; 8) Optional Safe Preview is activated upon request.

G. Architectural Properties

The given architecture has a number of main properties: Modularity, Scalability, Real-Time Capability, Hybrid Robustness, and User-Centric Design.

H. Architectural Significance

As opposed to traditional detection systems that are either isolated or run offline, the proposed architecture incorporates detection into the browsing process of the user. The system offers an all-encompassing and deployable answer to curb social media frauds.

V. IMPLEMENTATION

In this section, *The Guardian* implementation will be presented with the focus on the combination of the heuristic and deep learning elements, real-time processing, and the user-oriented safety mechanisms.

A. System Overview

The system is implemented as a hybrid client-server architecture consisting of a browser extension and a backend API. The design prioritizes real-time detection, low latency, and practical usability in social media environments.

B. Dataset Composition

To train and evaluate the AI model, six diverse text datasets were consolidated, yielding a total of 28,436 training samples. The dataset represents real-world scenarios and is naturally imbalanced, comprising 24,970 Safe samples (87.8%) and 3,466 Scam samples (12.2%). The malicious samples include social engineering attempts, cryptocurrency scams, and general phishing text. To manage the 7.2:1 class imbalance, a Weighted Cross-Entropy Loss function was applied during training. For the heuristic URL engine, a whitelist of 50,000 trusted domains (Tranco Top 50k) and a deduplicated blacklist of 1,981,138 malicious URLs were utilized to establish deterministic baselines.

C. Data Preprocessing and Tokenization

An important point of the implementation is the preprocessing strategy of textual data.

1) *URL Tokenization*: All the URLs are substituted by a token: instead of entering them directly to the model, they are substituted with a placeholder token.

$$T' = \text{replace}(\text{URL}, [\text{URL}]) \quad (9)$$

This approach prevents the model from memorizing specific malicious URLs and instead encourages learning of contextual scam patterns.

D. Implementation of AI Model

A fine-tuned DistilBERT model is used to implement the contextual analysis module.

- **Framework**: HuggingFace Transformers using PyTorch.
- **Task**: Binary classification (scam vs non-scam).
- **Output**: Probability score (S_{text}).

The base model utilized is `distilbert-base-uncased` (110M parameters). It was fine-tuned for 3 epochs using the AdamW optimizer with a learning rate of 5e-5, a batch size of 16, and a maximum sequence length of 128 tokens. To deal with the 7.2:1 class imbalance, a Weighted Cross-Entropy Loss function was applied, penalizing the misclassification of scams 7.2 times more than safe content to maximize recall.

E. URL Engine: Heuristic

The URL analysis engine is implemented as a lightweight rule-based system.

1) *Feature-Based Detection*: The engine checks URLs according to:

- Suspicious lexical patterns
- Use of IP addresses
- URL shortening services
- Domain anomalies

2) *Scoring Mechanism*: All the identified features add up to a risk score (S_{url}) resulting in rapid detection without model inference.

F. Hybrid Fusion Implementation

The system makes use of the maximum-risk fusion strategy:

$$R = \max(S_{url}, S_{text}) \quad (10)$$

This design will make sure that powerful markers of either of the components are maintained. This method (as opposed to averaging-based methods) does not dilute high-risk signals, making it more effective at finding adversarial or partially obfuscated scams.

G. Processing Pipeline and Backend API

FastAPI is used to implement the backend and is the heart of processing.

1) API Endpoints:

- `/scan`: Does hybrid scam detection.
- `/safe-preview`: Generates secure preview.
- `/proxy`: Allows restricted access to content.

2) *Processing Flow*: The flow includes input validation and preprocessing, parallel execution of URL and text analysis, risk score computation, and response generation.

H. Browser Extension Implementation

The browser extension has the responsibility of interacting with the web content in real-time. It identifies hyperlinks, extracts surrounding text, and makes structured requests to the backend. It provides feedback through risk levels (Safe, Suspicious, Dangerous) and facilitates the Safe Preview feature.

I. Safe Preview System

Safe Preview is a multi-layered security feature:

- 1) **Metadata Analysis**: Gets domain information, page title, and headers.
- 2) **Static Rendering**: Shows content but does not run any scripts.
- 3) **Sandbox Execution**: Isolates interaction to a controlled environment.

J. Real-Time Workflow

The system works in real-time via the following steps: 1) User accesses a webpage; 2) Extension extracts URL and context; 3) Information is transmitted to the backend; 4) URL and AI analysis are carried out; 5) Final risk is computed by fusion logic; 6) Results are shown immediately.

K. Implementation Summary

This implementation is a combined solution of hybrid detection, contextual learning, and real-time interaction. The system is practical and secure through effective heuristic analysis, transformer-based models, and the Safe Preview mechanism.

VI. RESULTS AND EVALUATION

In this section, the analysis of the proposed hybrid scam detection system performance will be measured based on the standard classification metrics. The assessment will be done on the overall performance and behavior in the classes with special emphasis on the effectiveness of scam detection.

A. Evaluation Setup

The model was tested on a labeled validation dataset of scam and non-scam samples. The measures of performance were standard measures of accuracy, precision, recall, and F1-score. To consider the imbalance of the classes, both the weighted and macro averages were taken into account.

B. Overall Performance

The overall results of the model were:

- Accuracy: 98.63%
- Precision (Weighted): 98.74%
- Recall (Weighted): 98.63%
- F1-Score (Weighted): 98.65%

As well, macro-averaged measures were:

- Precision (Macro): 95.16%
- Recall (Macro): 98.84%
- F1-Score (Macro): 96.90%

These findings show that the overall performance is strong, and there is a great level of consistency between majority and minority classes.

C. Class-Wise Performance

The close examination of the metrics by classes indicates the effectiveness of the model in identifying scam-related content.

1) Safe Class (Non-Scam):

- Precision: 99.88%
- Recall: 98.56%
- F1-Score: 99.21%

The accuracy is high, which means that valid content is barely recognized as a scam, reducing the number of unnecessary alerts.

2) Scam Class (Critical Class):

- Precision: 90.45%
- Recall: 99.13%
- F1-Score: 94.59%

The recall rate of 99.13% indicates that the system effectively captures the vast majority of scam instances, which is essential in security systems.

D. Confusion Matrix Analysis

The results of the confusion matrix are as follows:

- True Positives (TP): 682
- True Negatives (TN): 4923
- False Positives (FP): 72
- False Negatives (FN): 6

The fact that few false negatives (6 out of 688 scam samples) are missed by the system is an indication that there are very few scams that are missed by the system. This is especially essential, as missed scams are the greatest threat to the users.

E. Security-Oriented Evaluation

In terms of cybersecurity, the most important statistic is the recall of the scam class. The proposed system has a Scam Recall of 99.13% and a False Negative rate of less than 1%. This shows that the system can identify almost every bad case, significantly reducing user exposure to scams.

F. Trade-Off Analysis

Although the system has a high recall, the false positives are moderate with 72 cases of authentic material falsely labeled as spam. This trade-off is tolerable in security-oriented systems, where it is better to report potentially harmless content than fail to detect malicious behavior.

G. Real-Time Performance and Latency

To validate the system's real-time capabilities, the computational latency of the detection pipeline was measured. The system's layered architecture ensures that computational resources are optimized, resulting in an average end-to-end scan time of 150 to 250 milliseconds per request.

TABLE I
SYSTEM LATENCY BY COMPONENT

Pipeline Component	Average Execution Time
Whitelist / Blacklist Lookup	< 1 ms ($O(1)$ set lookup)
URL Heuristic Analysis	< 10 ms (CPU-bound)
AI Text Inference	100 – 200 ms
Total End-to-End Scan	150 – 250 ms

This sub-second processing speed ensures that users receive immediate visual feedback via the browser extension without experiencing noticeable delays during standard web browsing.

H. Comparative Analysis

To further assess the functionality of the proposed approach, a comparative analysis is done with representative baseline methods widely applied in scam detection. The baseline performance values are derived from representative results reported in prior studies [6], [7], [13], [15].

TABLE II
PERFORMANCE COMPARISON WITH BASELINE METHODS

Model	Accuracy	Precision	Recall (Scam)	F1-Score
URL-Based Heuristic	91.2%	89.5%	85.3%	87.3%
Traditional ML Model	94.8%	93.2%	91.0%	92.1%
Deep Learning (BERT)	97.2%	96.5%	95.8%	96.1%
The Guardian (Hybrid)	98.63%	98.74%	99.13%	98.65%

The findings show that the suggested hybrid solution is superior to single detection techniques. Specifically, the scam recall of 99.13% shows that the method of malicious content detection is much more advanced than the baseline models.

I. Discussion

The findings point to several strengths: 1) High Detection Capability; 2) Balanced Performance; 3) Effectiveness of Hybrid Approach; and 4) Minimal False Negatives.

J. Summary

The analysis demonstrates that the proposed hybrid framework achieves high accuracy and robustness. The system works particularly well on detecting malicious content with low false negatives.

VII. CONCLUSION AND FUTURE WORK

This paper proposed *The Guardian*, a hybrid architecture of real-time scam detection of social media users, which combines both heuristic URL-based and deep learning context-based understanding. The proposed system can be seen as a solution to the main shortcomings of current solutions, in that it brings structural and contextual analysis together, in a single, browser-based setting.

The hybrid detection method allows identifying explicit and implicit signs of scams efficiently based on the lightweight heuristic engine to perform rapid analysis of URLs and a DistilBERT-based model to classify data based on context. Therefore, the maximum-risk fusion mechanism is introduced to make sure that the critical risk signals are maintained and the robustness to detect in adversarial and obfuscated situations is enhanced. Also, the Safe Preview system offers a user-friendly level of security, which enables users to interact with suspicious content safely and minimize the exposure to possible threats.

Experimental evaluation shows that the proposed system achieves strong performance with a general accuracy of 98.63% and a recall of scam detection of 99.13%. The low false negative rates underscore the effectiveness of the system in reducing false threats and this is critical in security-based applications. These findings confirm the usefulness of the integration of the heuristic and deep learning methods in scam detection in real-time.

The suggested architecture also shows the realistic use of it through its usage as a browser extension with a FastAPI-backed backend, which is able to be easily integrated into the user environment. This fills the gap between the theoretical studies and implementation cybersecurity solutions.

A. Future Work

Although the suggested system exhibits good performance, there are some areas that can be improved:

- 1) **Multimodal Scam Detection:** Expand the system to examine images, videos and multimedia materials that are frequently employed in social media scams.
- 2) **Adaptive Learning Mechanisms:** Initiate progressive learning or online updating methods to be able to keep up with changing scam patterns and new threats.
- 3) **Cross-Platform Integration:** Increase the number of deployments on mobile apps and messaging platforms as well as browser extensions.
- 4) **Explainable AI (XAI):** Improve interpretability by explaining model decisions in detail, to enhance the user trust and transparency.

- 5) **Scalability and Distributed Deployment:** Scale back-end infrastructure to large scale deployment and real-time processing with high volume.

Overall, the proposed framework demonstrates the effectiveness of hybrid intelligence in advancing real-time cybersecurity solutions. *The Guardian* shows that integrating structural and contextual analysis within a user-centric system significantly enhances scam detection capabilities in modern social media environments. The framework establishes a strong foundation for future research in scalable, adaptive, and deployable security systems.

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